A SELF-STUDY TEXTBOOK · PRINT-READY

Linear Regression

A complete beginner-to-practitioner course covering intuition, mathematics, implementation, evaluation, assumptions, diagnostics, and the full practical ML workflow — built for projects, interviews, and real data analysis.

33 Chapters · Worked Examples · Interactive Visualizations · Interview Question Bank · Glossary + Cheat Sheet

CHAPTER 0

How to Use This Course

A short orientation before you begin — what you'll know by the end, and how to move through the material without getting stuck memorizing formulas.

Linear Regression is usually the very first machine learning algorithm anyone learns — and also the one people understand most shallowly, because it looks deceptively simple. This course is built to close that gap. By the end, you will be able to explain *why* the line sits where it sits, not just how to call `.fit()`.

What you will know by the end

You'll move through five layers for every idea: intuition, a tiny numeric example, the formula, a deeper explanation, and how it's actually used in a real ML project. By Chapter 32 you should be able to derive a regression line by hand, read a residual plot, explain assumption violations, and answer interview questions at beginner through advanced level.

Recommended learning sequence

Read chapters in order the first time — later chapters lean on earlier vocabulary (residual, SSE, OLS) constantly. Do the worked examples with pen and paper before checking the answer. Use the interactive visualizations to build intuition, not to skip the math.

> UNDERSTANDING VS. MEMORIZING
>
> **These are not the same skill**
>
> Memorizing $b_1 = \Sigma(x-\bar{x})(y-\bar{y}) / \Sigma(x-\bar{x})^2$ lets you pass a quiz. Understanding *why* that ratio produces the slope that minimizes squared error lets you reason about a model that's behaving strangely at 2am before a deadline. This course is built for the second kind of knowledge.

IF YOU REMEMBER ONLY 5 THINGS...

1. Linear Regression finds the single line that minimizes total **squared** residual error — that's what "best fit" means.
2. A residual is `Actual − Predicted`. Everything (SSE, MSE, RMSE, R^2) is built out of residuals.
3. R^2 compares your model to the simplest possible baseline: always predicting the mean.
4. Linear Regression assumes a roughly linear relationship, independent errors, constant error spread, and (for multiple features) low redundancy between predictors.
5. A single accuracy number can hide a badly-behaved model — always look at the residual plot.

The Big Picture

Beginner

What problem is Linear Regression actually solving, in plain language, before a single formula appears?

What problem are we solving?

Imagine you have this tiny dataset:

Hours Studied	Exam Score
1	45
2	50
3	55
4	65
5	70

THE QUESTION

Can we use the number of hours studied to **predict** exam score — even for a student whose hours we've never seen before, like 3.5 hours?

Notice the pattern: as hours go up, score tends to go up too, in a roughly steady way. Linear Regression is the tool for turning "roughly steady pattern" into a precise mathematical line you can use for prediction.

Input X → Model → Prediction ŷ

You feed in an input (hours studied), the model applies a learned rule, and it outputs a prediction (the little hat over ŷ signals "this is a prediction, not an observed fact").

Vocabulary you'll use constantly

Term	Meaning
Feature	An input variable used to make the prediction (e.g. hours studied)
Target	The value you're trying to predict (e.g. exam score)
Observation	One row of data — one student, one house, one transaction
Prediction ($\hat{y}$)	What the model outputs for a given input
Model	The learned rule that maps features to a prediction

Regression vs. Classification

Regression predicts a **continuous number**. Classification predicts a **category**.

Regression examples

- House price
- Salary
- Sales next month
- Temperature tomorrow

Classification examples

- Spam vs. not spam
- Churn vs. not churn
- Fraud vs. not fraud

REMEMBER THIS

If the answer to "what am I predicting?" is a number that could take any value on a continuum (₹52,340, 71.2°F, 3.8 kg), you're doing regression. If it's a label from a fixed set of categories, you're doing classification.